Preface to the Classics Edition

I am honored that SIAM is publishing my book in their Classics series. In preparing this new edition, I have significantly revised the portions of the book most in need of updating, primarily the pointers to available software and the bibliography. The bulk of the content of the book has aged well, however, and remains largely intact. I like to think that this is because of the book's modern perspective, but it may be due more to its foundational nature, primarily covering the most basic concepts and methods of numerical analysis and scientific computing, which change relatively slowly (Newton, Euler, and Gauss haven't published much lately!). Perhaps the most significant technical change is the adoption of a barycentric formulation for Lagrange polynomial interpolation, which had previously appeared only in an exercise.

A number of new trends have emerged since the original publication of the book, some of which are now alluded to in this new edition, while others are beyond its scope. From a pedagogical perspective, perhaps the most significant trend has been the emergence of Python as a programming language and computing environment that has had notable impact on both education and professional practice. Accordingly, pointers to appropriate Python software have been added to the software citations in the book. Another notable new development on the software front is `Chebfun`, which combines several traditional threads in innovative ways to provide insightful new computing capabilities. The trend toward ever greater exploitation of parallelism in large-scale scientific computing has continued apace, but serious coverage of parallel algorithms would require another book.

Another trend is the increasing use of stochastic methods, for example in linear algebra and optimization, often motivated by applications in rapidly growing areas such as machine learning and data science. The book already contained a brief

chapter on stochastic methods that provides preparation for further study from other sources. Interest in uncertainty quantification has also surged in recent years. With its consistent emphasis on existence, uniqueness, conditioning, and stability, the book lays a solid conceptual foundation for assessing the trustworthiness of computational results, but again detailed coverage of the latest UQ techniques is beyond our scope.

A new development since the publication of the previous edition is the availability of about 100 educational modules that I have created in collaboration with a number of students. These Java applets dynamically illustrate many of the concepts and methods of scientific computing and serve as interactive examples to accompany the book. They are suitable for classroom demonstrations by instructors or for self-study by students. The modules can be accessed via a web browser, or a stand-alone version can be downloaded and run locally on the user's computer. The modules can be freely obtained at `www.siam.org/books/cl80`.

In addition to the two mentors (Alston Householder and Gene Golub) that I acknowledged in the original preface, I would like to take this opportunity to acknowledge the influence of several authors whose careful and appealing writing styles I have found consistently inspiring and worthy of emulation: Nick Higham, Pete Stewart, Gil Strang, Nick Trefethen, and Charlie Van Loan. Finally, I want to acknowledge once again the patience and understanding of my wife, Mona, as I continue to work despite "retirement" — it is fitting that with this new edition she finally gets her own dedication page!

Michael T. Heath